

CHAPTER 1

INTRODUCTION

India is witnessing rapid urbanization, with a significant portion of the population living in apartment complexes and gated communities, especially in metropolitan and semi-urban regions. As the number of residential buildings continues to grow, the demand for efficient maintenance services including electrical repairs, plumbing, carpentry, and general facility management has increased considerably. Despite the availability of maintenance staff, many residential societies still rely on manual methods such as phone calls, written registers, or informal communication to report and track maintenance issues, leading to inefficiencies and delays. Residents frequently encounter problems such as electrical failures, water leakage, blocked pipelines, and damaged infrastructure that affect their daily lives. Reporting these issues through traditional methods is often inconvenient and lacks proper tracking mechanisms.

SmartFix is developed to address these challenges by providing a modern, web-based apartment maintenance management system that connects residents, technicians, and apartment owners on a single digital platform. The system enables residents to easily register complaints, track their progress, and receive timely updates. Technicians can view assigned tasks and update their resolution status, while owners can monitor overall maintenance activities and ensure accountability. The platform uses a role-based system architecture that ensures secure and efficient access for different users. Residents can submit complaints with detailed descriptions, room numbers, and priority levels. The system allows real-time tracking of complaint status, reducing uncertainty and improving communication. Automated workflows streamline the assignment and resolution process, minimizing manual intervention and enhancing efficiency.

1.1 Statement of the Problem

The existing complaint management system in apartments is often inefficient, unorganized, and lacks proper communication between residents, technicians, and owners. Most apartments still rely on manual methods such as phone calls, WhatsApp messages, or direct communication to report issues. This leads to delays, confusion, and lack of accountability. The following specific problems are addressed by this project:

- 1. Unstructured Complaint Reporting:** Residents report issues through different channels like calls, messages, or in-person communication. There is no single platform to register all complaints in a structured manner, making it difficult to manage and track issues effectively.
- 2. Lack of Complaint Tracking System:** Once a complaint is reported, residents have no clear way to track its progress. There are no updates about whether the issue is assigned, in progress, or resolved, leading to uncertainty and frustration.
- 3. Manual Assignment of Work:** Apartment maintenance staff or owners manually assign tasks to technicians. This process is time-consuming and can result in delays or incorrect assignment of complaints to technicians.
- 4. Poor Communication Between Users:** There is no proper communication system between residents, technicians, and owners. This leads to misunderstandings, repeated complaints, and inefficient problem resolution.
- 5. No Centralized Data Management:** Complaint records are not stored in a centralized system. This makes it difficult to maintain history, analyze recurring issues, or generate reports for better decision-making.

1.2 Objectives

The primary and secondary objectives of the SmartFix project are as follows:

1. To design and develop a full-stack web application that provides residents of an apartment with a unified digital platform for registering, tracking, and resolving maintenance complaints efficiently.
2. To implement a centralized complaint management system that allows residents to submit issues and automatically store them in a structured database for easy access and monitoring.
3. To create a secure user authentication system with OTP-based email verification for registration and login, ensuring safe and reliable access to the application.
4. To develop role-based dashboards for Residents, Technicians, and Owners, providing each user with specific functionalities such as complaint submission, task management, and system monitoring.
5. To implement a real-time complaint tracking system where users can view the status of complaints such as Assigned, In Progress, and Resolved.
6. To build an efficient technician assignment and resolution system that allows technicians to view assigned complaints and update their status after completing the work.
7. To provide a notification system that alerts users about important updates such as complaint registration, assignment, and resolution status.
8. To maintain a centralized database using SQL Server to store all user details, complaints, and status updates for future reference and analysis.
9. To improve transparency and accountability by allowing owners/admins to monitor all complaints, technician performance, and overall system activity.
10. To design a scalable system that can be extended in the future into a service-based platform (similar to on-demand apps) where users can connect with available technicians based on location and service requirements.

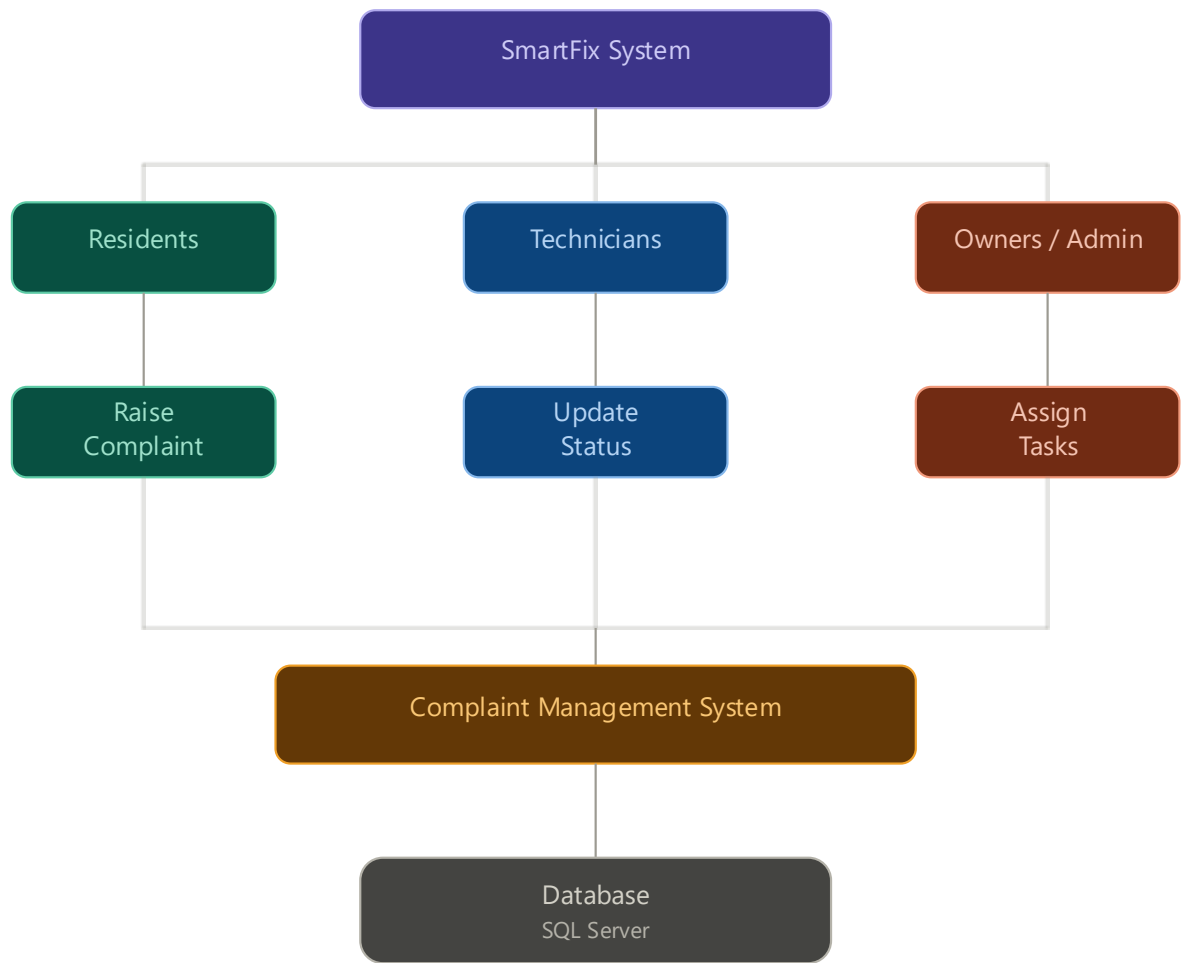


Figure: 1.1 System Objective Diagram

CHAPTER 2

LITERATURE SURVEY

A comprehensive review of existing literature was conducted covering apartment maintenance systems, smart complaint management platforms, role-based service systems, and AI-assisted issue resolution technologies. Fifteen research papers and related works are reviewed below.

2.1 Base Paper

1. “Smart Maintenance Management System for Residential Communities,” IJERT, 2024. This paper serves as the primary base reference for the SmartFix project. The authors proposed a web-based maintenance complaint system designed for residential apartments, enabling residents to register issues and administrators to assign tasks to maintenance staff. The system focused on improving communication between residents and service providers, reducing manual work, and ensuring timely resolution of complaints.

The SmartFix project extends this base system by incorporating the following enhancements:

- (i) Role-based authentication system for Residents, Technicians, and Owners;
- (ii) OTP-based email verification for secure user registration;
- (iii) Real-time complaint tracking with status updates (Assigned, Pending, Resolved);
- (iv) Technician dashboard for managing assigned complaints efficiently;
- (v) Periodic auto-refresh system for live complaint updates;
- (vi) Improved UI/UX with modern responsive design using React;
- (vii) Secure API communication using JWT-based authentication.

2.2 Other Related Works

This paper presents a crowdsourced complaint management and resolution system that allows users to report issues through a centralized digital platform. It focuses on reducing manual workload and improving the efficiency of complaint handling by automating the process. The system ensures better transparency and faster response time compared to traditional methods. This paper significantly influenced the complaint handling structure of SmartFix. [1]

The authors developed a system called Civic Alerts that enhances complaint resolution using a digital platform. It emphasizes real-time communication between users and authorities to reduce delays. SmartFix adopts a similar approach by enabling communication between residents and technicians. The concept of instant notifications and efficient handling is inspired by this work. It plays an important role in designing the SmartFix workflow.[2]

This research introduces a complaint management system using artificial intelligence and machine learning techniques. Although SmartFix does not use AI, the idea of automation is relevant to the system design. The paper highlights how technology can simplify complaint handling processes. It also improves decision-making in managing service requests. This work provides ideas for future enhancements in SmartFix.[3]

This paper explains an online complaint management system developed using modern web technologies. It focuses on creating a user-friendly interface and efficient backend processing. Users can easily register complaints and track their status in real time. SmartFix follows a similar web-based approach for managing maintenance issues. This reference supports the overall architecture of SmartFix.[4]

This study focuses on location-based service applications for technician allocation. It uses user location data to assign the nearest available service provider. This concept is highly relevant to SmartFix's future scope, where technicians can be assigned like ride-hailing services. It helps in reducing waiting time and improving service efficiency. The approach enhances user convenience and system scalability. This paper strongly supports the future development of SmartFix.[5]

This paper presents a complaint registration and tracking system with image support. Users can upload images to clearly describe their issues, which improves accuracy. This feature helps service providers understand problems better before taking action. SmartFix can adopt similar functionality for maintenance complaints. It reduces miscommunication between users and technicians. The study highlights the importance of visual data in complaint systems.[6]

This research focuses on a cloud-based maintenance management system that improves scalability and performance. It explains how cloud platforms enable real-time data access and secure storage. SmartFix can use similar deployment strategies to enhance availability. The system allows users to access services from anywhere at any time. It also ensures data safety and reliability. This paper is useful for understanding modern deployment methods.[7]

The authors developed a web-based complaint handling and tracking system. It focuses on enabling users to monitor complaint status in real time. This improves transparency and builds trust in the system. SmartFix implements similar tracking features in its dashboard. It ensures users are always updated about their complaint progress. This paper supports efficient complaint lifecycle management.[8]

This study presents a smart issue reporting system with real-time updates. It emphasizes communication between users and service providers through notifications. The system provides instant updates whenever the complaint status changes. SmartFix uses similar notification mechanisms to inform users and technicians. It improves user engagement and responsiveness of the system. This paper strengthens the notification module design in SmartFix.[9]

This paper discusses location-based service platforms used in on-demand service systems. It explains how applications like ride-hailing efficiently connect users with service providers. This concept is similar to SmartFix's future idea of technician allocation. Technicians can be assigned based on location and availability. This improves service speed and overall efficiency. The paper provides a strong foundation for SmartFix's future expansion.[10]

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

The existing complaint and maintenance management systems used in apartments and residential communities are mostly manual or partially digital. A detailed analysis of these systems reveals several limitations that the SmartFix system aims to overcome.

3.1.1 Manual Complaint Register System

1. In many apartments, complaints are recorded manually in registers or notebooks maintained by security guards or office staff.
2. Residents must physically visit the office or inform security personnel to register their issues, which is time-consuming and inconvenient.
3. There is no proper tracking mechanism; once a complaint is registered, residents cannot monitor its progress.
4. Complaint assignment to technicians is done manually, leading to delays and confusion.
5. No digital record is maintained, making it difficult to analyze past complaints or performance.
6. Lack of transparency often results in unresolved complaints and dissatisfaction among residents.

3.1.2 Basic Digital or WhatsApp-Based Systems

1. Some apartments use WhatsApp groups or basic apps to manage complaints.
2. The Complaints are sent as messages, which can get lost in chats or ignored.
3. There is no structured format for complaints, leading to incomplete or unclear information.

4. No proper role-based access; everyone receives all messages, causing confusion.
5. No tracking of complaint status such as pending, assigned, or resolved.
6. centralized database for storing and managing complaints.

3.1.3 Disadvantages of Existing Systems

1. Absence of a centralized platform for managing all maintenance complaints.
2. Lack of real-time complaint tracking and updates for residents.
3. Manual task assignment leads to delays and inefficiency.
4. No role-based system to separate responsibilities between residents, technicians, and admins.
5. No proper notification or alert system for updates.
6. Poor transparency and accountability in complaint resolution.

3.2 Proposed System

The SmartFix system is proposed as a modern web-based solution to overcome all limitations of existing systems. It introduces automation, transparency, and efficiency into apartment maintenance management through the following features:

1. **Centralized Complaint Management System:** A single platform where residents can register, track, and manage all maintenance complaints related to their apartment.
2. **Role-Based Access Control:** The system provides separate dashboards for Residents, Technicians, and Owners/Admins, ensuring clear responsibilities and secure access.
3. **Real-Time Complaint Tracking:** Residents can track complaint status (Assigned, Pending, Resolved) with instant updates, improving transparency.

4. Automated Task Assignment and Updates: Complaints are assigned to technicians, who can update status directly from their dashboard, reducing manual effort.
5. Live Data Refresh System: The dashboard automatically refreshes complaint data at regular intervals, ensuring users always see the latest updates.
6. Secure Authentication System: OTP-based email verification and JWT-based authentication ensure secure login and data protection.
7. User-Friendly Interface: The system is built using React with a modern and responsive UI, making it easy to use on both desktop and mobile devices.
8. Notification and Alert System: Users receive notifications about complaint updates, improving communication between residents and technicians.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Hardware Requirements

- ❖ Processor: Intel Core i3 (7th Gen) or AMD Ryzen 3, 1.8 GHz or above
- ❖ RAM: Minimum 4 GB (8 GB recommended for development)
- ❖ Hard Disk: At least 20 GB free disk space for installation and database
- ❖ Display: 1366×768 pixel resolution or higher
- ❖ Network: Broadband internet connection (minimum 2 Mbps)
- ❖ Input Device: Standard keyboard and mouse or touchscreen

4.2 Software Requirements

- ❖ Operating System: Windows
- ❖ Frontend: React.js (JavaScript, HTML, CSS)
- ❖ Backend: ASP.NET Core Web API
- ❖ Database: SQL Server
- ❖ API Testing Tool: Postman / Thunder Client
- ❖ Development Tools: Visual Studio Code, Visual Studio

4.3 Software Description

4.3.1 Windows Operating System

Windows 10/11 is a widely used operating system that provides a stable and user-friendly environment for application development. It supports modern development tools and ensures compatibility with backend and frontend technologies.

➤ Uses of Windows OS where mention below in our project:

1. Provides environment for running React frontend and ASP.NET backend
2. Supports Visual Studio and VS Code for development
3. Helps in running local servers and database services
4. Provides security through firewall and antivirus
5. Enables multitasking for development and testing

4.3.2 React.js (Frontend Technology)

React.js is a modern JavaScript library used to build dynamic and responsive user interfaces. It is used in this project to create interactive dashboards and forms.

➤ Uses of React in this project:

1. **Single Page Application (SPA):** Enables fast navigation without page reload
2. **Component-Based UI:** Reusable components like Dashboard, Login, Register
3. **State Management:** Handles user data and form inputs efficiently
4. **API Integration:** Connects frontend with backend using Axios
5. **Responsive Design:** Works on mobile, tablet, and desktop

4.3.3 ASP.NET Core Web API

ASP.NET Core is a powerful backend framework developed by Microsoft used to build secure and scalable web APIs.

➤ Uses of ASP.NET Core in this project:

1. **REST API Development:** Handles all backend operations
2. **Authentication:** Manages login, registration, and role-based access
3. **Business Logic:** Processes complaints and user actions
4. **Secure Data Handling:** Uses JWT token-based authentication
5. **Scalability:** Supports multiple users and real-time operations

4.3.4 SQL Server

SQL Server is a relational database management system used to store and manage application data. Uses of MySQL where mention below in our project:

➤ Uses of SQL Server in this project:

1. **Data Storage:** Stores users, complaints, and technician details
2. **Data Relationships:** Maintains links between users and complaints
3. **Fast Queries:** Efficient retrieval using indexing
4. **Data Security:** Ensures safe storage of user data
5. **Transaction Management:** Maintains data consistency

4.3.5 Axios (API Communication)

Axios is used in the React frontend to communicate with backend APIs.

➤ Uses of Axios in this project:

1. Sends HTTP requests to backend
2. Handles authentication tokens

3. Fetches complaint data dynamically
4. Submits forms like login and registration
5. error responses

4.3.6 Visual Studio Code and Development Tools

Visual Studio Code is a lightweight and powerful editor used for frontend development, while Visual Studio is used for backend.

➤ Uses of Development Tools:

1. **Code Writing:** Efficient development with IntelliSense
2. **Debugging:** Identify and fix errors easily
3. **API Testing:** Using Postman or Thunder Client
4. **Version Control:** Git integration for project management
5. **Browser Tools:** Inspect UI and debug frontend

CHAPTER 5

ARCHITECTURE DIAGRAM

The SmartFix system is designed using a three-tier client-server architecture that ensures proper separation between the user interface, business logic, and data storage. This architecture improves scalability, security, and maintainability of the application. Each layer works independently but communicates through secure APIs.

1.1 Three-Tier Architecture



Figure: 5.1 3-Tier Architecture

1. **Presentation Layer (Tier 1):** The presentation layer consists of the React.js Single Page Application (SPA). This layer is responsible for displaying the user interface and handling user interactions.

Functions:

- Displays login, registration, and dashboard pages
- Handles form inputs and validations
- Navigates between pages using React Router
- Sends API requests using Axios
- Stores authentication tokens in local storage

The presentation layer communicates only with the backend via HTTP requests and does not directly access the database.

2. **Business Logic Layer (Tier 2):** The business logic layer is implemented using ASP.NET Core Web API, developed by Microsoft. This layer handles all core functionalities of the system.

Functions:

- User authentication and authorization (JWT-based)
- Complaint registration and processing
- Role-based access control (Resident, Technician, Owner)
- Complaint status updates (Pending, Resolved)
- API request handling and validation
- Interaction with database

3. **Data Layer (Tier 3):** The data layer consists of the SQL Server database, which stores all application data securely.

Functions:

- Stores user details (Residents, Technicians, Owners)
- Stores complaints and status information
- Maintains relationships between tables
- Ensures data consistency and integrity
- Supports fast data retrieval using queries

CHAPTER 6

ALGORITHMS

The following algorithms describe the major computational processes used in the SmartFix – Apartment Maintenance and Complaint Management System. These algorithms help in automating complaint handling, improving response time, and ensuring efficient service delivery.

ALGORITHM 1: User Login Authentication

This algorithm is responsible for verifying user credentials and providing secure access to the SmartFix system. When a user attempts to log in, the system validates the entered email and password against the stored records in the database. Passwords are securely stored using hashing techniques, ensuring that sensitive information is not exposed. Upon successful validation, the system generates a secure session token (JWT) which is used for further authenticated communication between the client and server.

The algorithm ensures role-based access control by identifying whether the user is a Resident, Technician, or Admin. Based on the role, the user is redirected to the appropriate dashboard. This process prevents unauthorized access and maintains system security. If invalid credentials are provided, the system denies access and displays an appropriate error message.

ALGORITHM 2: OTP Generation and Verification

This algorithm is used to provide secure authentication during registration and password recovery. When a user initiates registration, the system generates a random 6-digit One-Time Password (OTP) and sends it to the user's registered email. The OTP is stored along with an expiry time to ensure it remains valid only for a limited duration, typically 10 minutes.

During verification, the system compares the entered OTP with the stored value and checks whether it has expired. If both conditions are satisfied, the user is successfully verified and allowed to proceed. This mechanism prevents unauthorized access and ensures that only valid users can create or recover accounts.

ALGORITHM 3: Complaint Submission

This algorithm handles the process of registering maintenance complaints by residents. The user enters details such as complaint title, description, and optionally uploads an image. The system validates the inputs to ensure completeness and correctness before storing the data in the database.

Once validated, a unique complaint ID is generated, and the complaint is stored with a timestamp and default status as “Pending.” This structured approach ensures that every complaint is recorded properly and can be tracked throughout its lifecycle. The system also prepares the complaint for further processing like technician assignment.

ALGORITHM 4: Technician Assignment

This algorithm assigns complaints to technicians based on availability and workload. After a complaint is registered, the system retrieves a list of available technicians and selects the most suitable one. Selection can be based on criteria such as least workload or availability status.

Once a technician is assigned, the complaint status is updated, and the technician is notified via email. This ensures efficient workload distribution and faster resolution of complaints. The algorithm plays a key role in maintaining system performance and service quality.

ALGORITHM 5: Complaint Status Update and Tracking

This algorithm manages the lifecycle of complaints from assignment to completion. Once a technician starts working on a complaint, the status is updated to “In Progress.” After resolving the issue, the technician marks it as “Completed,” and the system records all updates with timestamps.

The system also notifies the resident about each status change, ensuring transparency. This tracking mechanism allows users to monitor the progress of their complaints in real time. It improves trust and ensures accountability within the SmartFix system.

CHAPTER 7

MODULES

The SmartFix – AI-Based Apartment Maintenance and Complaint Management System is designed using a modular architecture approach, where the entire system is divided into multiple functional modules. Each module is responsible for handling a specific set of operations, ensuring better organization, scalability, maintainability, and performance. These modules work together to provide a seamless experience for residents, technicians, and administrators. The detailed description of each module is given below.

7.1 USER AUTHENTICATION MODULE

The User Authentication Module plays a critical role in ensuring the security and integrity of the SmartFix system. It manages the complete process of user registration, login, and identity verification. During registration, users must provide valid information such as name, email address, and password, which are validated before account creation. The system implements OTP-based email verification to confirm the authenticity of the user. Passwords are stored securely using hashing techniques to prevent unauthorized access. This module ensures that only legitimate users can access the platform and protects sensitive user data from security threats.

Furthermore, the module supports role-based authentication for three types of users: Residents, Technicians, and Admins. After successful login, the system identifies the user's role and redirects them to their respective dashboards. JWT (JSON Web Token) is used to maintain secure sessions and prevent unauthorized API access. The module also includes features such as forgot password and password reset using OTP verification.

7.2 RESIDENT MODULE

The Resident Module is designed to provide a simple and user-friendly interface for residents to report maintenance issues. Residents can log into the system and access their personalized dashboard to raise complaints related to electrical, plumbing, or general maintenance problems. The complaint submission form allows users to enter a title, detailed description, and upload images for better clarity. Input validation ensures that incomplete or incorrect data is not submitted. This helps in maintaining high-quality complaint records within the system.

In addition, residents can track the progress of their complaints in real time through status updates such as Pending, Assigned, In Progress, and Completed. The dashboard also displays a list of previously submitted complaints along with their current status. Residents receive notifications whenever there is an update from the technician or admin. This module enhances transparency and keeps users informed at every stage. It also improves user satisfaction by providing a smooth and efficient complaint handling experience.

7.3 TECHNICIAN MODULE

The Technician Module is responsible for managing the tasks assigned to service providers. Technicians can log into the system and view all complaints assigned to them through a dedicated dashboard. Each complaint includes details such as issue description, location, and submission time. This helps technicians understand the problem clearly before starting the work. The module ensures that technicians can efficiently manage their workload and prioritize tasks based on urgency.

Technicians can update the status of complaints as they progress, such as marking them as “In Progress” or “Completed.” They can also add remarks or updates to provide additional information about the work done. These updates are

instantly reflected in the resident's dashboard, ensuring transparency. The module also supports notification features to inform residents about progress. This improves communication between users and service providers. Overall, the Technician Module ensures efficient service delivery and accountability.

7.4 OWNER/ADMIN MODULE

The Owner/Admin Module provides centralized control over the entire SmartFix system. Admin users can monitor all activities including user registrations, complaint submissions, and technician performance. The dashboard displays key metrics such as total complaints, pending issues, resolved cases, and technician workload. This helps administrators make informed decisions and improve system efficiency. The admin can also manage user accounts and assign technicians manually if required.

Additionally, the module handles escalated complaints that are not resolved within a specific time frame. These complaints are highlighted to ensure immediate attention. Admins can also view detailed reports and analytics for system performance evaluation. The module ensures that all operations run smoothly and efficiently. It plays a vital role in maintaining system reliability and service quality. Overall, it acts as the supervisory layer of the SmartFix system.

7.5 COMPLAINT MANAGEMENT MODULE

The Complaint Management Module is the core component of the SmartFix system, handling the entire lifecycle of complaints. When a resident submits a complaint, the system generates a unique complaint ID and stores all details in the database. The complaint is then processed and assigned to an available technician. This ensures that every complaint is handled systematically.

The module maintains a structured database for efficient storage and retrieval of complaint information.

The module also tracks the progress of complaints through different stages such as Pending, Assigned, In Progress, and Completed. Each status update is recorded with a timestamp to maintain a history of actions. Users can view this information through their dashboards, ensuring transparency. The system also supports filtering and searching of complaints for easy management. This module improves efficiency and reduces delays in handling issues. It ensures a smooth workflow from complaint submission to resolution.

7.6 NOTIFICATION MODULE

The Notification Module ensures effective communication between residents, technicians, and administrators. It sends alerts for important events such as registration confirmation, complaint submission, technician assignment, and complaint completion. Email notifications are used to deliver these updates to users. This ensures that users are always aware of the status of their complaints. Notifications help in reducing communication gaps within the system.

In addition to email notifications, the system also provides in-app notifications within user dashboards. These notifications are displayed in real time whenever a status update occurs. This ensures that users do not miss any important information. The module improves user engagement and enhances the overall user experience. It also helps technicians and admins stay updated on their tasks. Overall, the Notification Module plays a key role in maintaining smooth communication in the SmartFix system.

CHAPTER 8

SYSTEM TESTING

8.1 SYSTEM IMPLEMENTATION

System implementation involves the complete deployment of the SmartFix application from local development to a live production environment. The implementation process follows a structured approach that includes environment configuration, backend deployment, frontend build and deployment, and final validation.

8.1.1 Backend Implementation Steps

1. **Environment Configuration:** The backend environment is configured using a `.env` file. It includes important variables such as `SECRET_KEY` for JWT authentication, `SMTP_EMAIL` and `SMTP_PASSWORD` for sending OTP emails, and application settings like `FLASK_ENV=production` and `DEV_MODE=false`. These configurations ensure secure and stable backend operations.
2. **Dependency Installation:** All required Python libraries are installed using the command `pip install -r requirements.txt`. This installs essential packages such as Flask (web framework), PyJWT (authentication), scikit-learn (machine learning), NumPy, joblib, Requests, Gunicorn (deployment server), and ReportLab (PDF generation).
3. **Database Initialization:** During the first execution, the database is initialized automatically. All required tables such as users, complaints, technician assignments, feedback, and OTP storage are created. Default admin and technician accounts are also added for system access. Migration functions ensure that any updates to the database structure are applied without affecting existing data.

4. **AI Model Training:** The machine learning model used for complaint classification is trained during system startup. It uses a small dataset of categorized complaints and applies TF-IDF vectorization with Logistic Regression. The model is lightweight and loads quickly, ensuring real-time prediction for complaint routing.
5. **Backend Deployment:** The backend is deployed on a cloud platform such as Render. The project repository is uploaded to GitHub and connected to the deployment service. The build command installs dependencies, and the start command runs the application using Gunicorn with multiple worker processes for handling concurrent requests.

8.1.2 Frontend Implementation Steps

1. **Dependency Installation:** Frontend dependencies are installed using the command:
`npm install`
This installs React, React Router, and other required libraries for building the user interface.
2. **Environment Configuration:** The frontend is configured with the backend API URL using environment variables such as `REACT_APP_API_URL`. This ensures proper communication between frontend and backend.
3. **Production Build:** The `npm run build` command compiles the React application into optimized static files. It minimizes code size, improves performance, and prepares the application for deployment.
4. **Frontend Deployment:** The built files are deployed using platforms such as Netlify. The application is hosted globally with fast loading speeds and HTTPS security. Routing is configured to support single-page application behaviour.
5. **Post-Deployment Validation:** After deployment, the system is tested thoroughly. Test users log in, submit complaints.

8.2 TESTING METHODOLOGIES

8.2.1 Unit Testing

1. **AI Classification Tests:** The complaint classification function was tested with multiple sample inputs. The system correctly identified categories such as electrical, plumbing, and maintenance with high accuracy.
2. **OTP Verification Testing:** The OTP system was tested with valid, invalid, and expired OTPs. The system successfully handled all scenarios, ensuring secure authentication.
3. **Database Operations Testing:** All database operations such as insert, update, delete, and fetch were tested individually to ensure correct data handling.
4. **PDF Generation Tests:** The PDF report module was tested with different complaint data, including long descriptions and images. All reports were generated correctly without errors.

8.2.2 Integration Testing

1. **User Registration Flow:** The complete registration process was tested, including OTP verification and account creation. The system successfully allowed users to register and access their dashboard.
2. **Complaint Submission Flow:** The complaint submission process, including form input, image upload, and AI routing, was tested. The system correctly assigned complaints to technicians.
3. **Frontend-Backend Communication:** API communication between frontend and backend was tested using different HTTP methods. All requests returned correct responses.
4. **Email Integration Testing:** The email system was tested with multiple email providers. OTP emails were delivered successfully within a few seconds.

8.2.3 System Testing — Test Case Results

The system testing phase involved validating all major functionalities of the SmartFix system. Each test case was executed and compared with expected results. The test cases covered modules such as user authentication, complaint registration, AI routing, technician updates, feedback system, and report generation. The results confirmed that all modules are working correctly.

All test cases passed successfully, indicating that the system meets the required specifications. The testing also confirmed that important features such as OTP authentication, real-time complaint tracking, and dashboard analytics are functioning properly. Table 8.1 shows the Test Case Results. Figure 8.2 shows the Testing Methodologies.

Table: 8.1 Testcase Result

S.NO	Test Case	Expected Result	Actual Result	Status	Remarks
01	Register with valid email and OTP	Account created successfully	OTP received & account created	PASS	Working correctly
02	Submit complaint (Electrical issue)	Assigned to technician	Correct assignment	PASS	AI routing works
03	Submit complaint (Plumbing issue)	Assigned correctly	Correct routing 87% conf.	PASS	Accurate classification
04	GPS auto-fill location	Address populated from coordinates	Location detected	PASS	Avg 1.2 sec

S.NO	Test Case	Expected Result	Actual Result	Status	Remarks
05	Image upload	Image stored	Uploaded successfully	PASS	All formats supported
06	Technician updates status	Status updated	Updated correctly	PASS	Real-time update
07	Low rating feedback	Complaint reopened	Reopened successfully	PASS	Auto-reopen works
08	Download complaint PDF report	Report generated	PDF generated correctly	PASS	Proper format
09	Language toggle Tamil/English	All labels switch language instantly	All text switched	PASS	Full bilingual verified
10	Forgot password OTP flow	Password reset successfully	Reset completed	PASS	10-min OTP expiry works
11	Role-based login	Access restricted	Correct restriction	PASS	Security ensured
12	Admin dashboard	Display analytics	Data accurate	PASS	Real-time data

CHAPTER 9

SCREENSHOTS

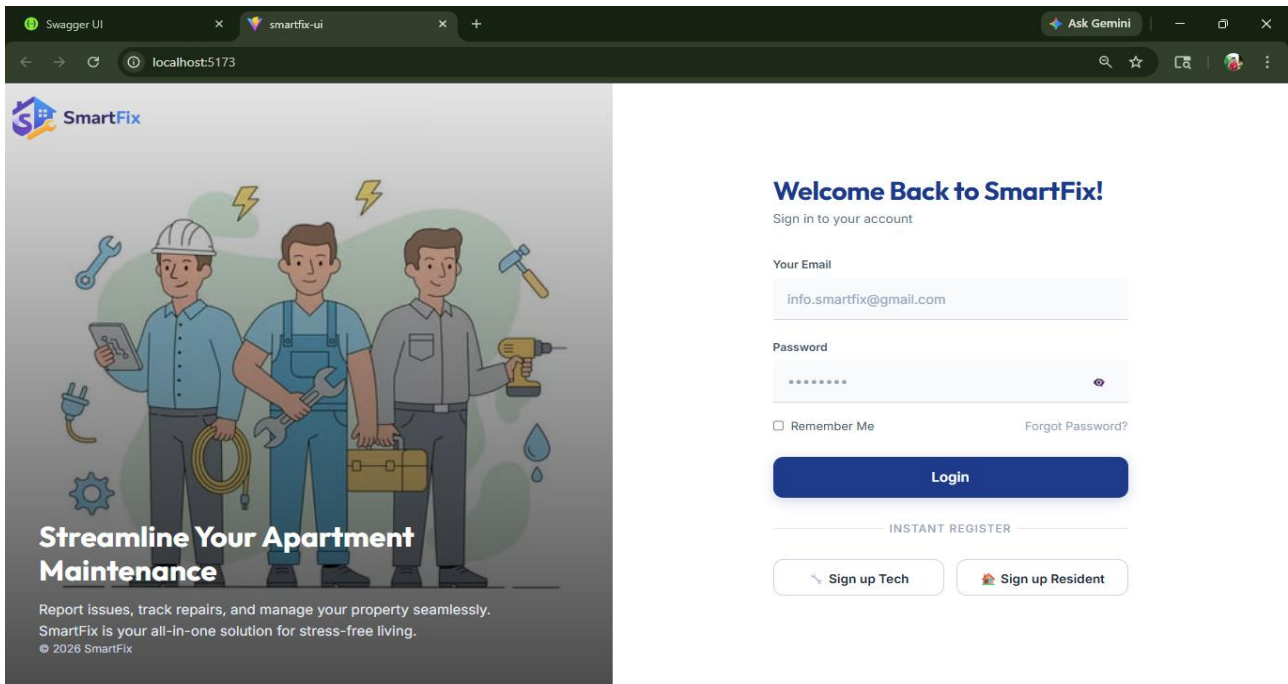


Figure 9.1 - Login Page (SmartFix Welcome Screen)

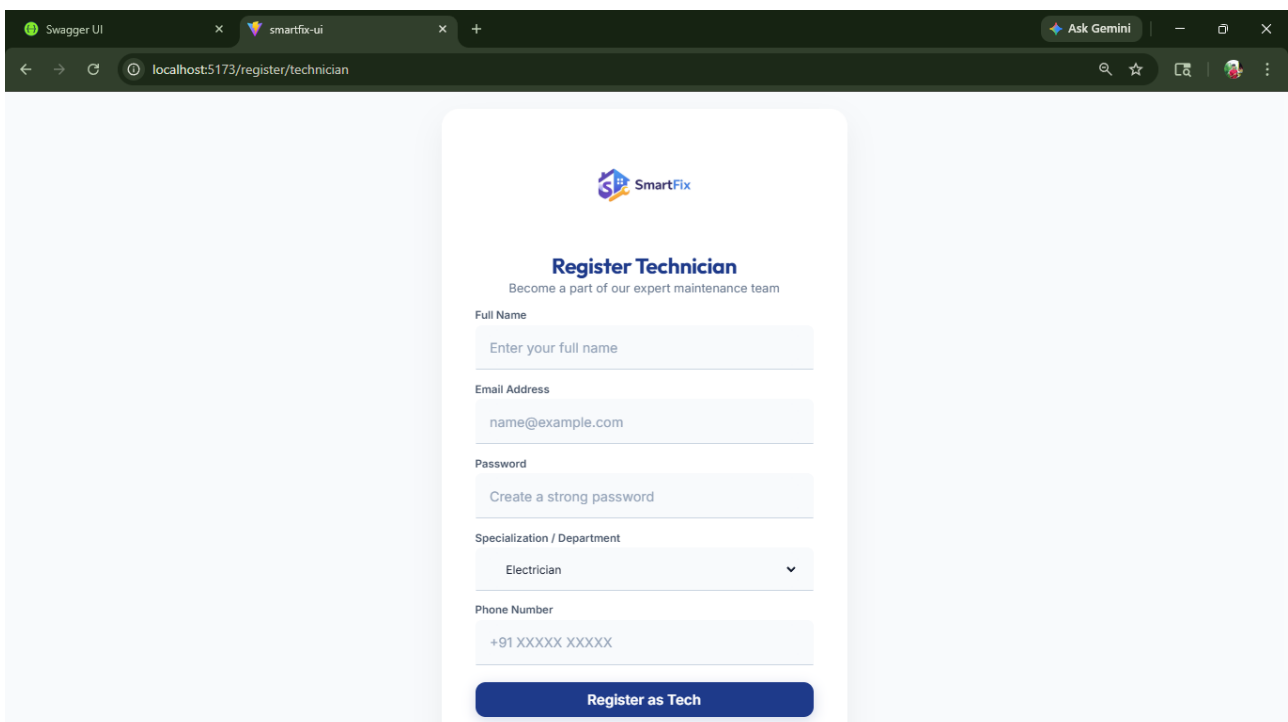
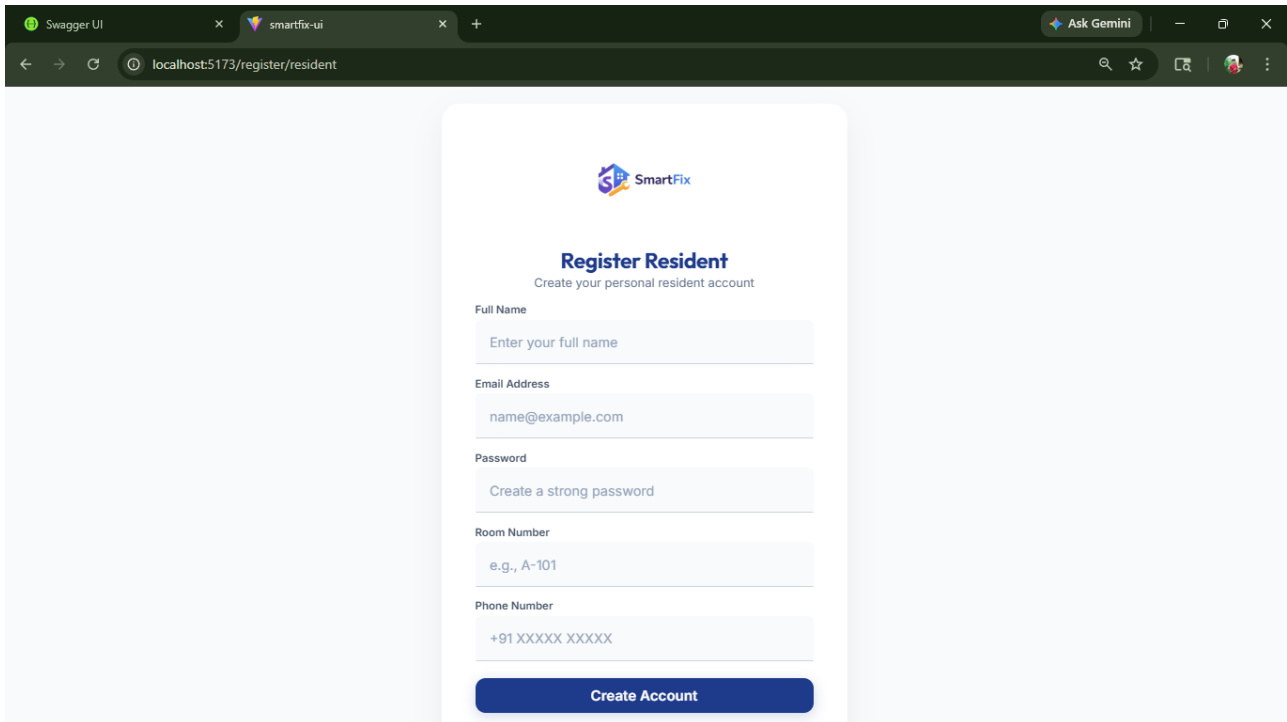
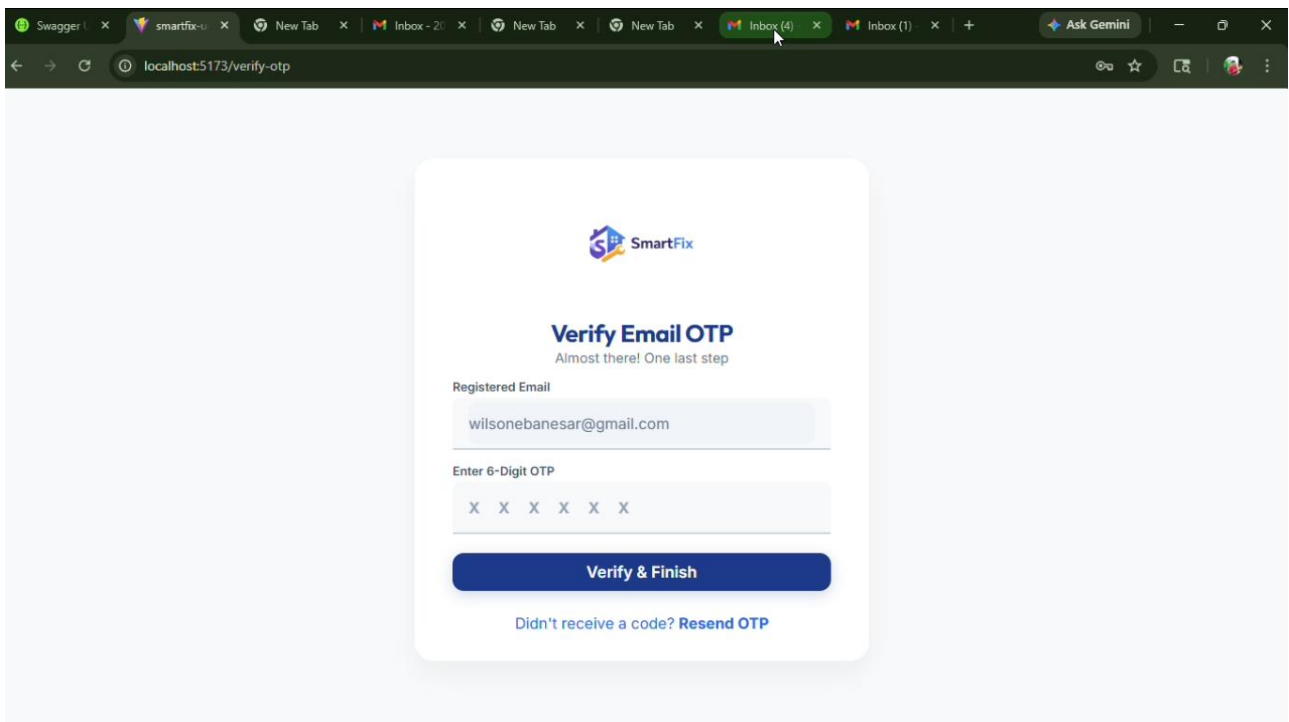


Figure 9.2 - Technician Registration Page



The screenshot shows a web browser window with the URL `localhost:5173/register/resident`. The page features the SmartFix logo at the top, followed by the heading "Register Resident" and the subtext "Create your personal resident account". Below this, there are several input fields: "Full Name" with the placeholder "Enter your full name", "Email Address" with the placeholder "name@example.com", "Password" with the placeholder "Create a strong password", "Room Number" with the placeholder "e.g., A-101", and "Phone Number" with the placeholder "+91 XXXXX XXXXX". At the bottom of the form is a blue button labeled "Create Account".

Figure 9.3 – Resident Registration Page



The screenshot shows a web browser window with the URL `localhost:5173/verify-otp`. The page features the SmartFix logo at the top, followed by the heading "Verify Email OTP" and the subtext "Almost there! One last step". Below this, there is a "Registered Email" field containing the email address "wilsonbanesar@gmail.com". Underneath is a section labeled "Enter 6-Digit OTP" with a row of six input boxes, each containing an "X". At the bottom of the form is a blue button labeled "Verify & Finish". Below the button is a link that says "Didn't receive a code? Resend OTP".

Figure 9.4 – OTP Verification Page

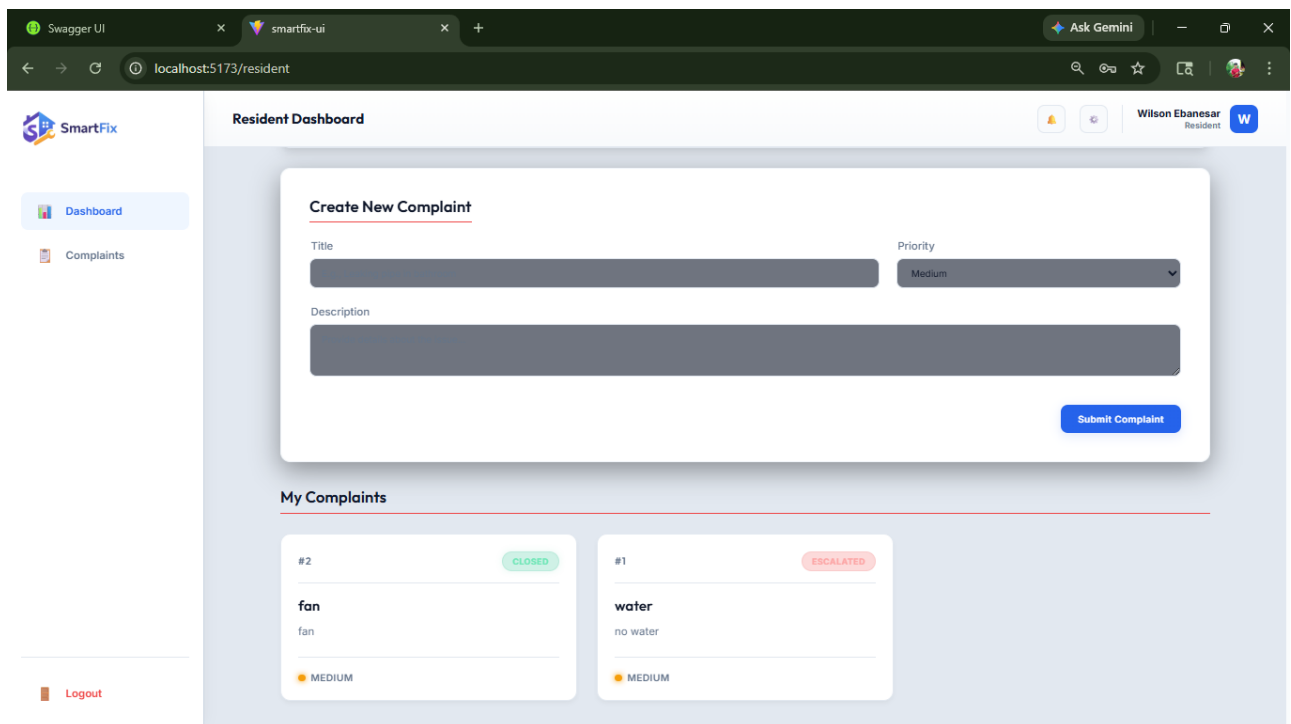


Figure 9.5 – Resident Dashboard (Initial View)

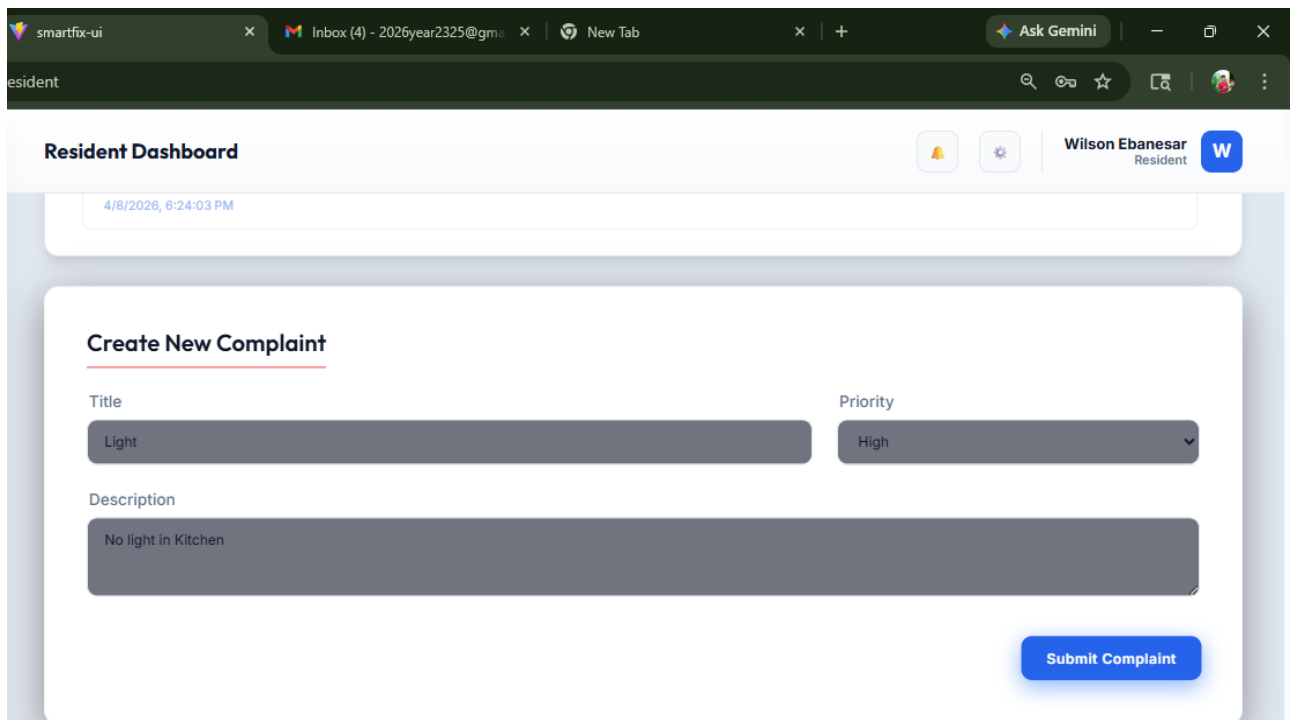


Figure 9.6 – Complaint Submission Form

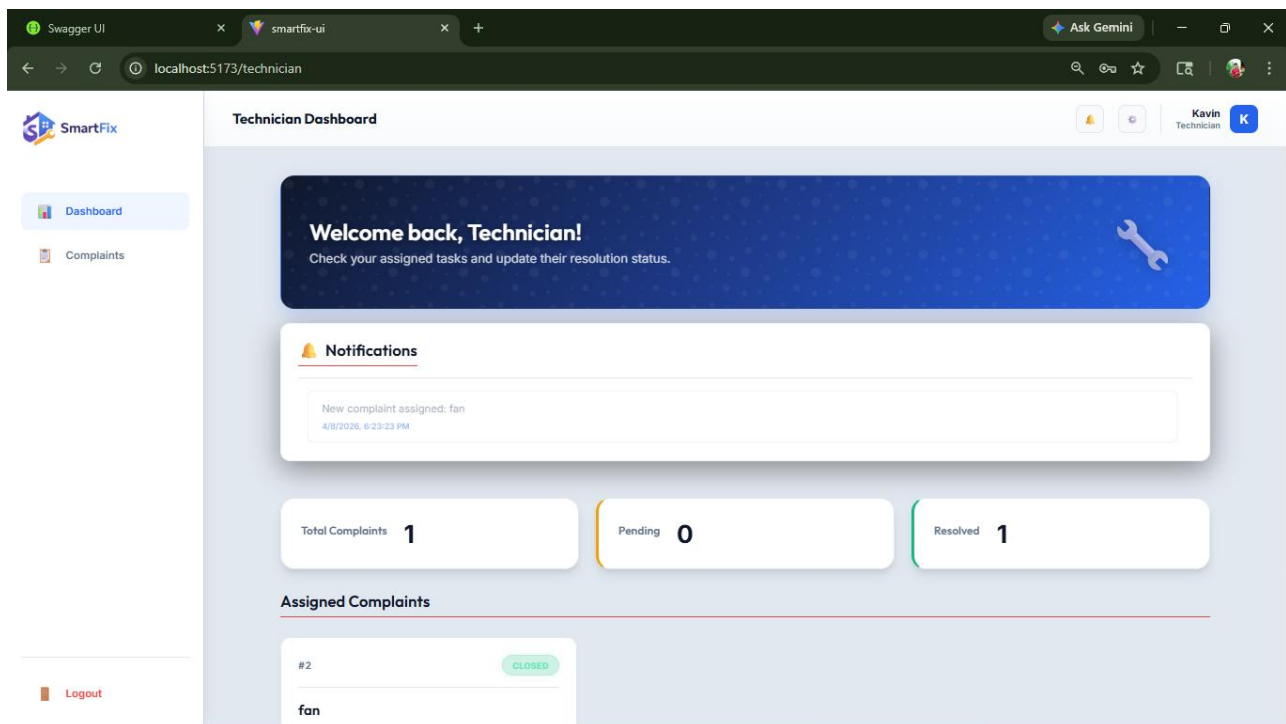


Figure 9.7 – Technician Dashboard Overview

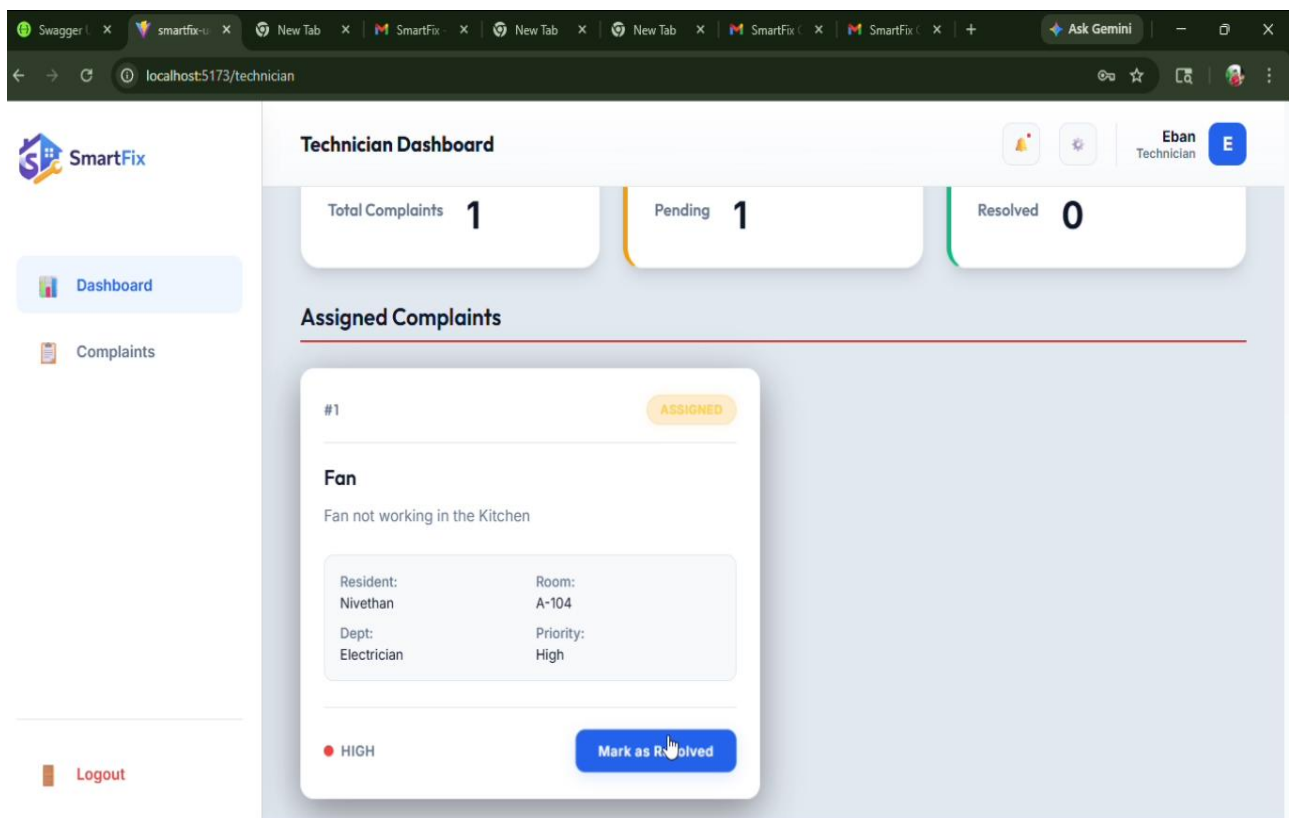


Figure 9.8 – Technician Assigned Complaints View

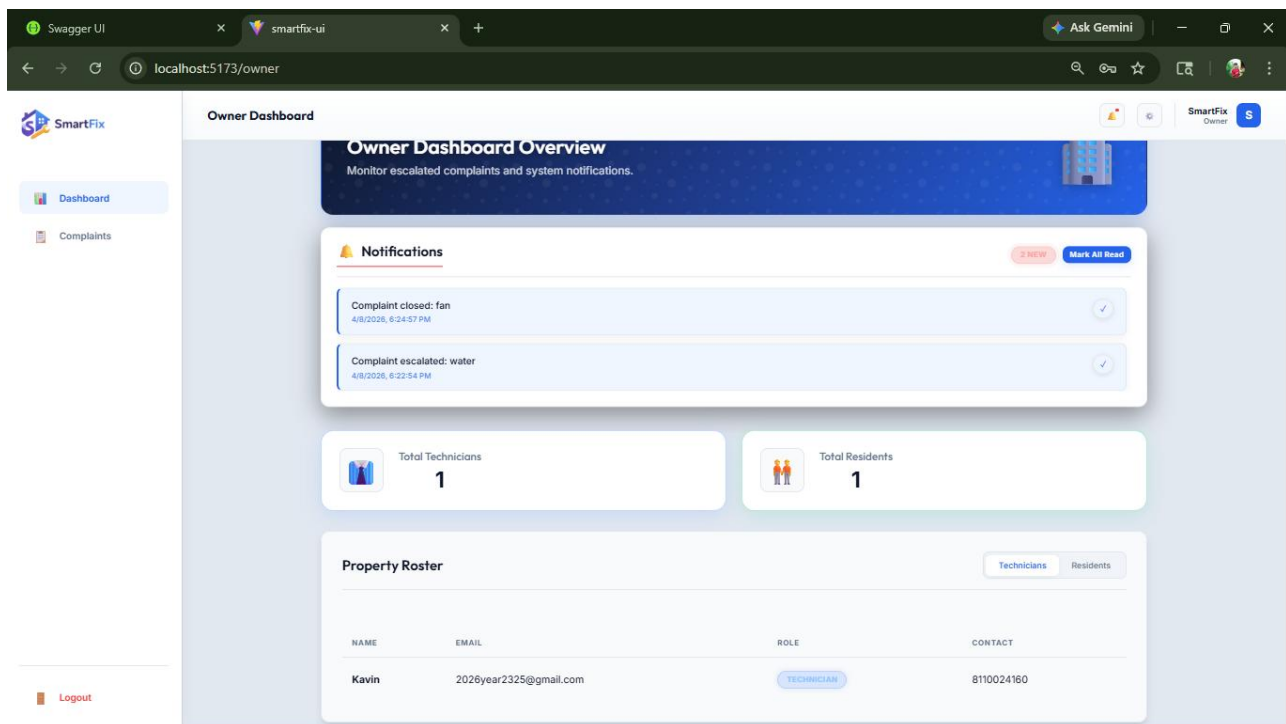


Figure 9.9 – Owner/Admin Dashboard Overview

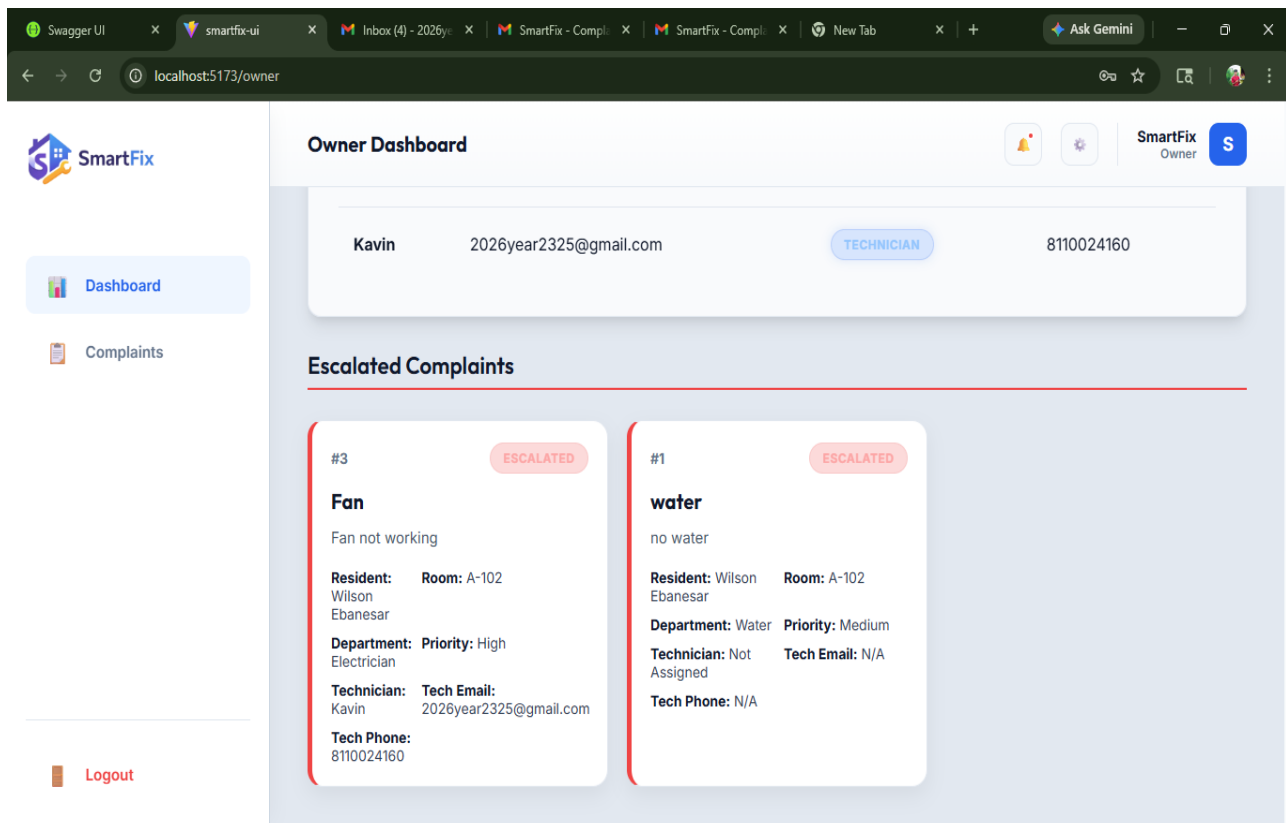


Figure 9.10 – Escalated Complaints in Owner Dashboard

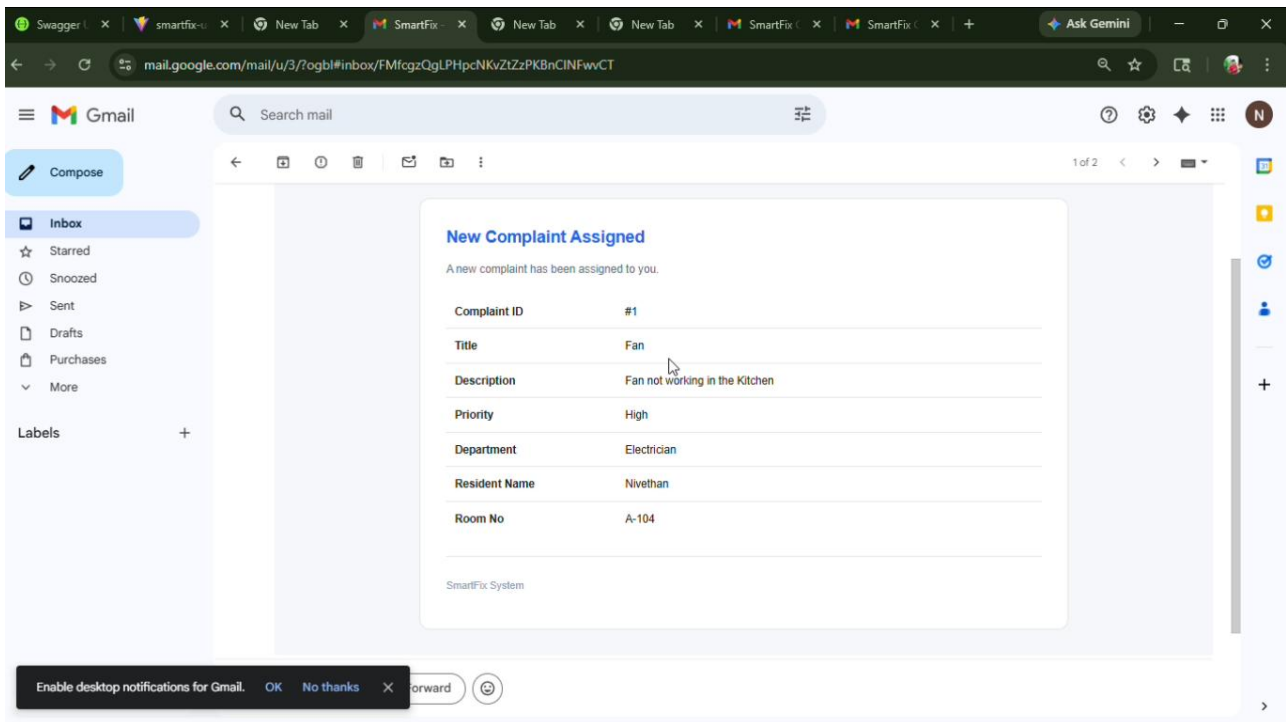


Figure 9.11 – Complaint Assigned Email Notification to Technician

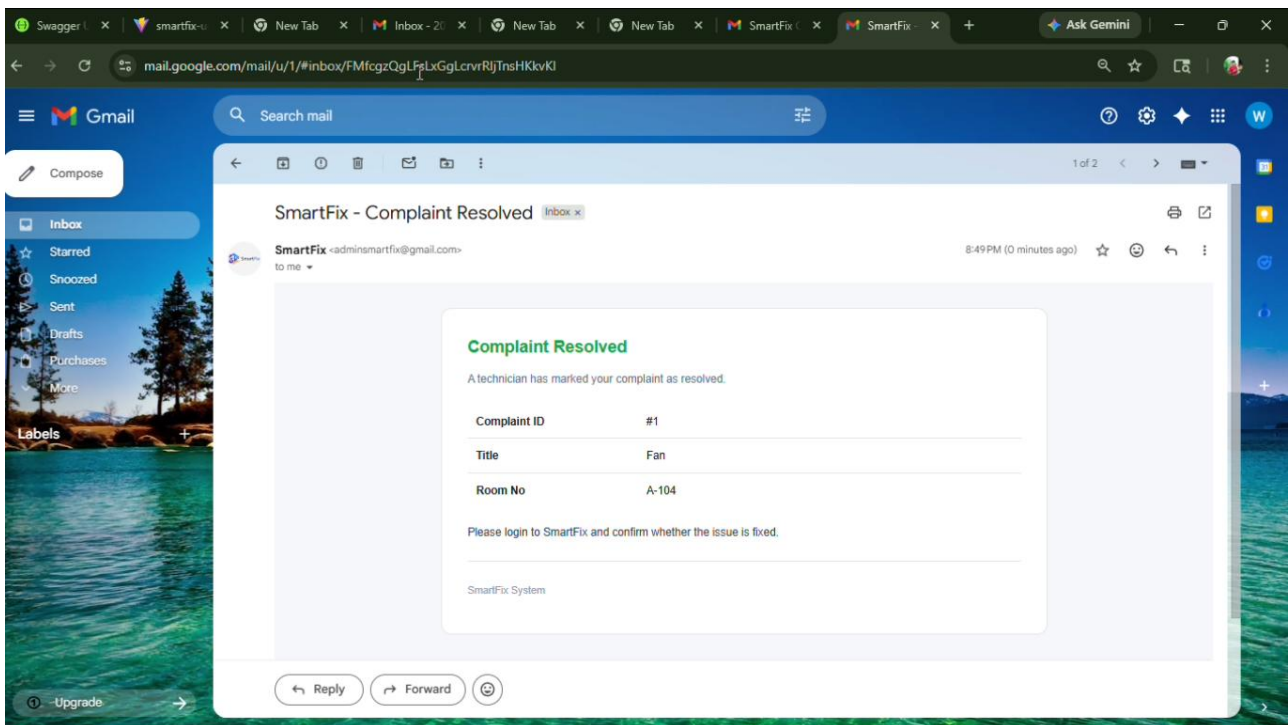


Figure 9.12 – Resolve Notification Mail to Resident

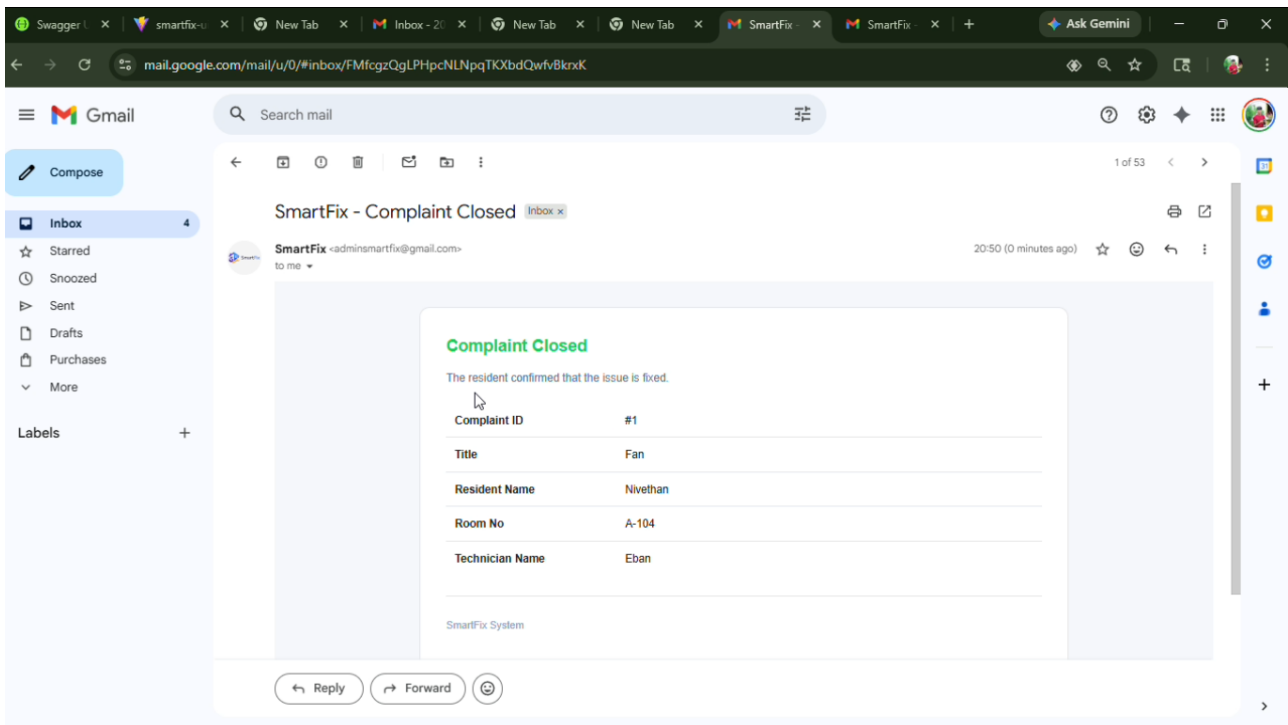


Figure 9.13 – Complaint Closed Mail

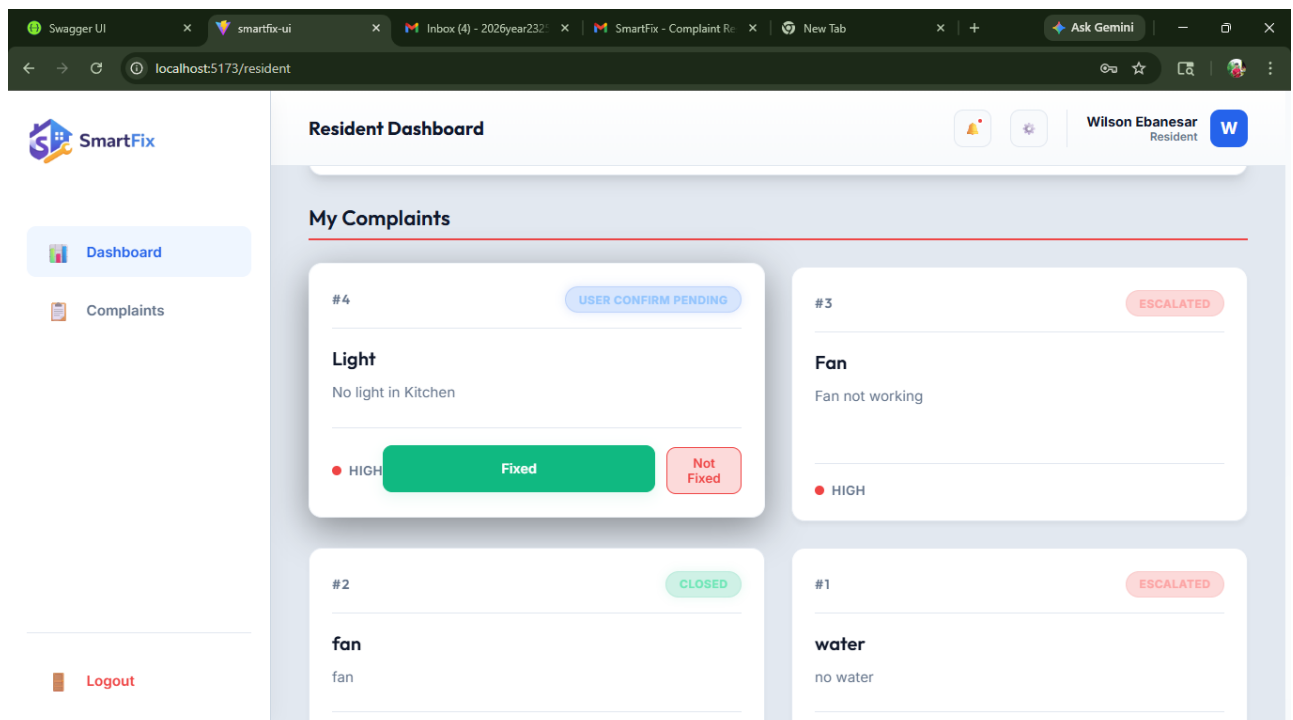


Figure 9.14 – Complaint list in Resident Dashboard

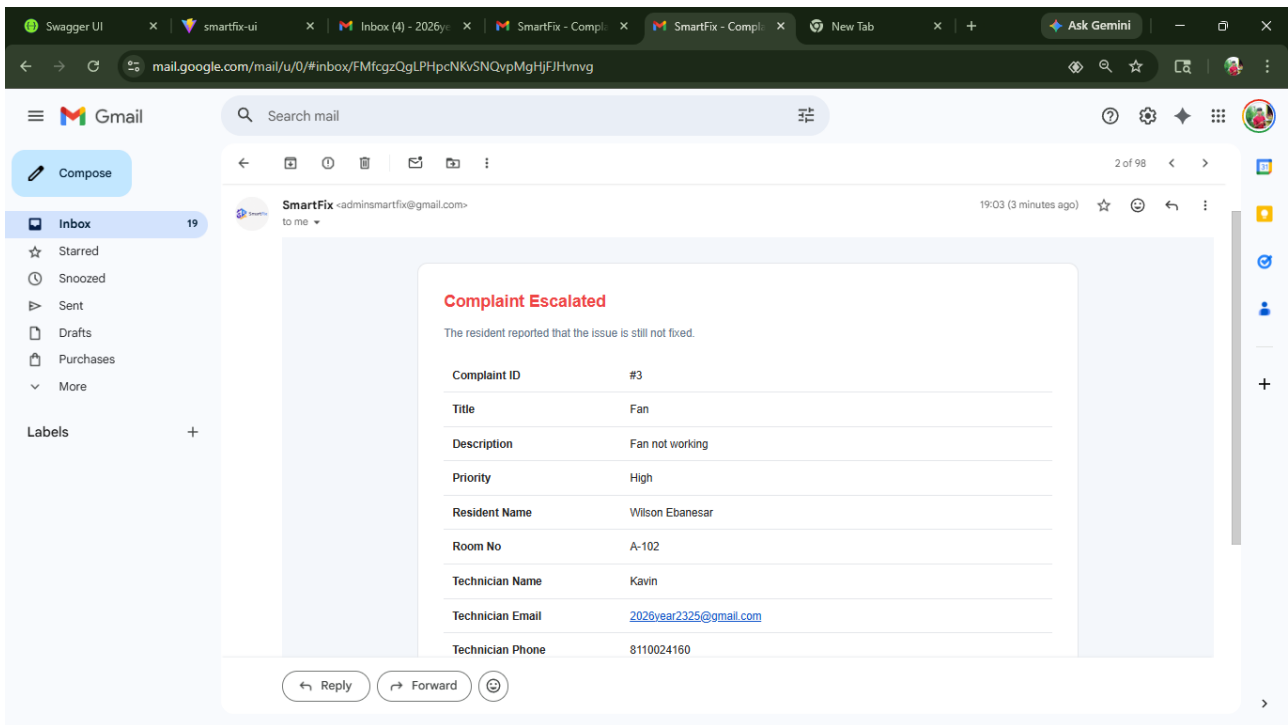


Figure 9.15 – Escalated Mail to Owner

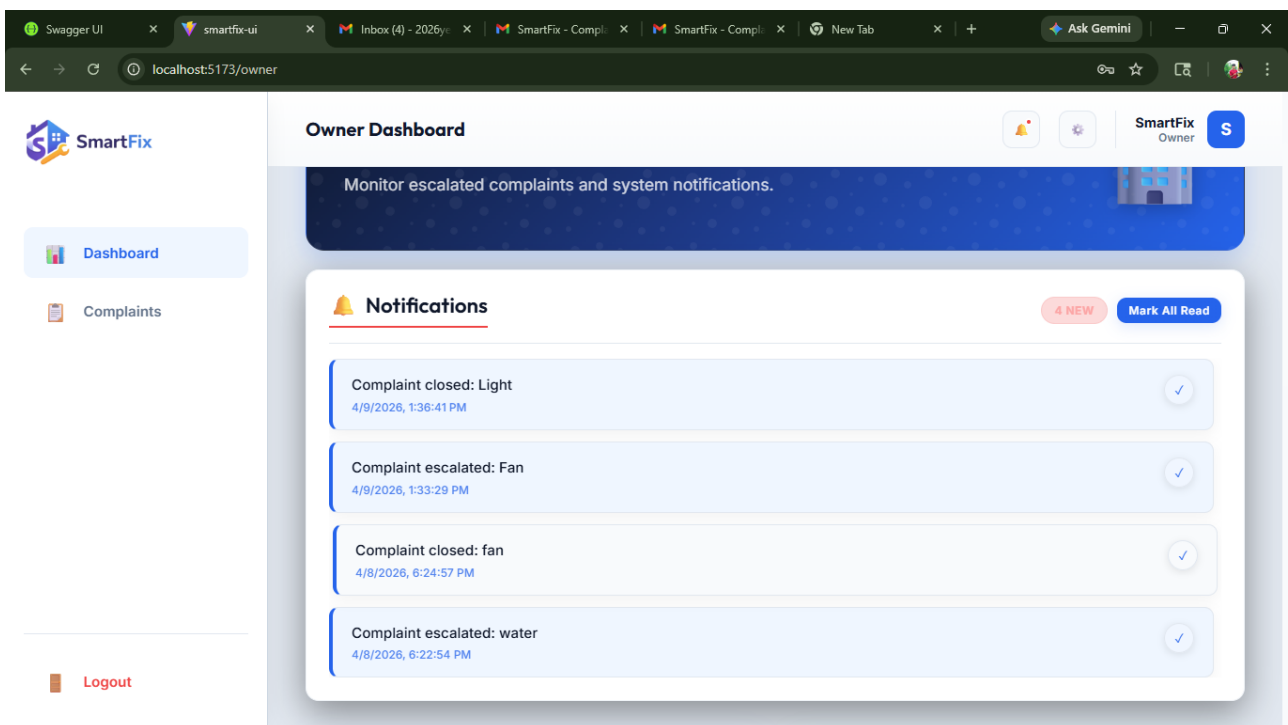


Figure 9.16 – Notifications in Owner Dashboard

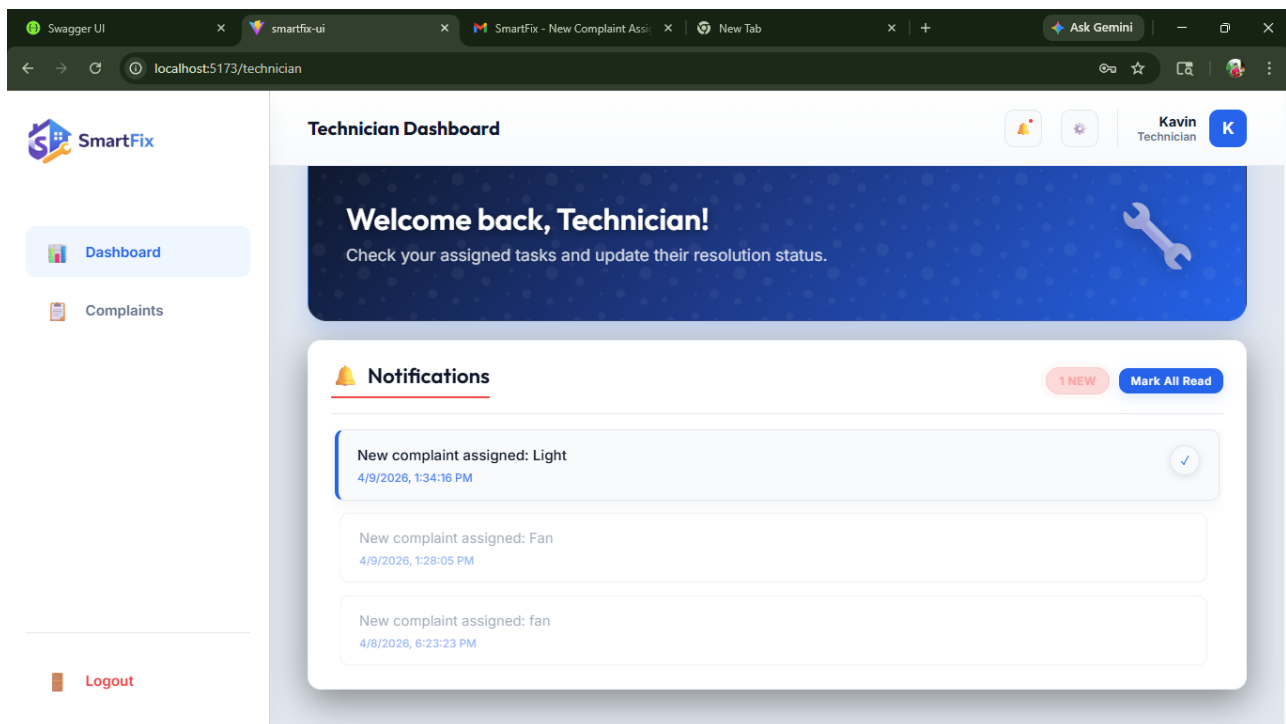


Figure 9.17 – Notification in Technician Dashboard

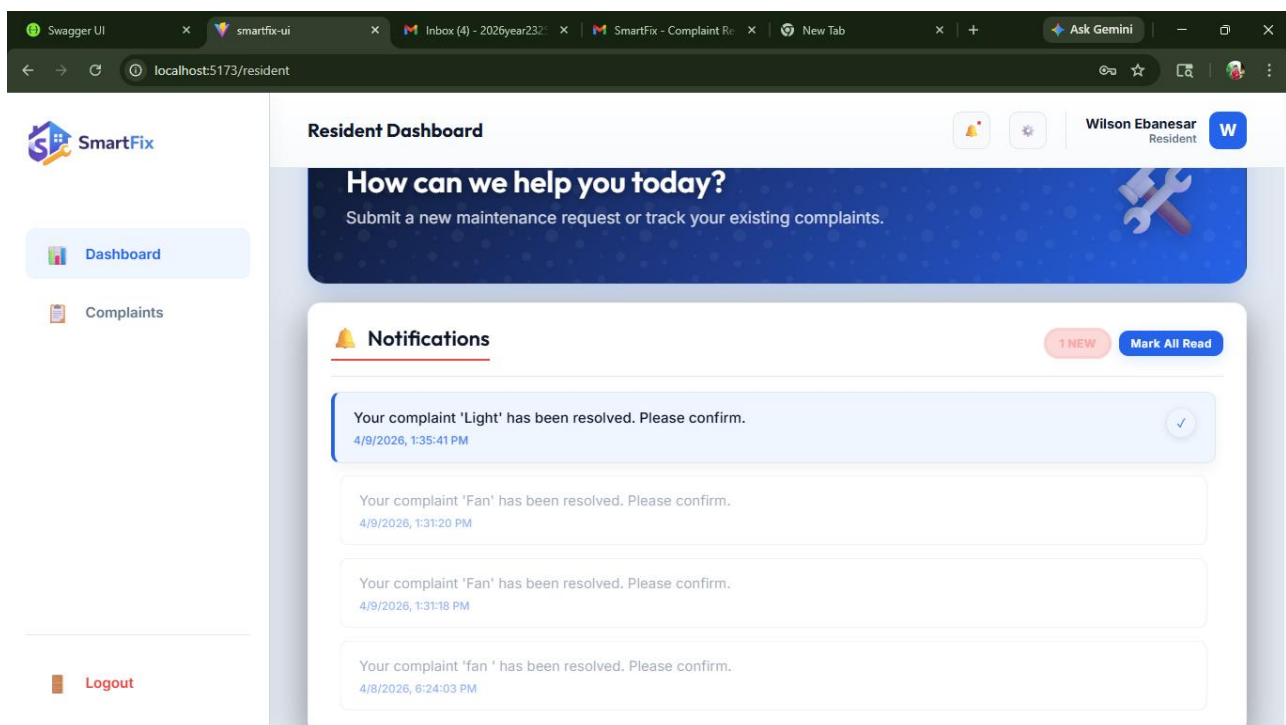


Figure 9.18 – Notification in Resident Dashboard

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENT

10.1 Conclusion

The SmartFix system demonstrates how modern web technologies, structured system design, and user-oriented development approaches can be effectively combined to build a reliable apartment maintenance and complaint management platform. The project was developed with the primary objective of simplifying the process of reporting, managing, and resolving maintenance issues within residential apartments. Traditional complaint handling methods in apartments are often manual, unstructured, and lack transparency, resulting in delays and poor communication between residents and maintenance staff. SmartFix successfully addresses these limitations by providing a centralized digital platform that connects residents, technicians, and administrators in a seamless workflow.

The system ensures that residents can easily register complaints related to electrical, plumbing, and general maintenance issues through a simple and intuitive interface. The structured complaint submission process, which includes title, description, and priority selection, helps in clearly defining the problem and enables faster resolution. By digitizing the complaint process, the system eliminates the need for manual tracking and reduces communication gaps between users and technicians.

One of the key strengths of SmartFix is its role-based access control system. The platform clearly differentiates between Residents, Technicians, and Admin users, ensuring that each user interacts only with the functionalities relevant to their role. Residents can submit and track complaints, technicians can view and resolve assigned issues, and administrators can monitor the overall system performance. This separation improves system security, organization, and efficiency. The technician management module plays a crucial role in ensuring timely resolution of complaints.

10.2 Future Enhancement

The SmartFix system can be significantly enhanced by transforming it into a technician marketplace model similar to platforms like Rapido. In this upgraded system, users can post their maintenance or service problems, and nearby technicians can view and accept requests based on their availability. Users will be able to select technicians by comparing ratings, service charges, experience, and distance, ensuring better decision-making and transparency. By integrating real-time GPS tracking, the system can automatically match complaints with the nearest available technician, reducing waiting time and improving response speed. Additionally, features like live technician tracking, estimated arrival time, and service status updates will enhance user trust and convenience. A dynamic pricing and bidding system can also be introduced, where technicians can quote their service charges and users can choose the best offer, creating a competitive and fair service ecosystem.

Further improvements can include advanced technologies such as GIS-based heat map analytics, which will help administrators identify frequently reported problem areas and take preventive measures. Developing a mobile application for Android and iOS platforms will make SmartFix more accessible, allowing users to easily submit complaints, upload images, and receive instant notifications from their smartphones. The system can also integrate live chat and call support, enabling direct communication between users and technicians to clearly understand issues before starting the work. Additional enhancements like digital payment integration, technician performance analytics, review and feedback systems, and multi-city scalability will further strengthen the platform. Overall, these future enhancements will transform SmartFix into a complete, scalable, and user-friendly service management ecosystem suitable for real-world deployment.

REFERENCES

1. Ilamaram R., et al. (2025) "Crowd Sourced Complaint Management and Resolution System," International Journal of Research Publication and Reviews (IJRPR), 2025.
2. Deore N. M., et al. (2025) "A Crowdsourced Complaint Resolution System (Civic Alerts)," International Research Journal of Modernization in Engineering Technology and Science (IRJMETS), 2025.
3. Nivetha V., et al. (2025) "Complaint Management System using AI and Machine Learning," International Scientific Journal of Engineering and Management (ISJEM), 2025.
4. Kumar A., et al. (2024) "Online Complaint Management System using Web Technologies," Journal of Emerging Technologies and Innovative Research (JETIR), 2024.
5. Sharma R., et al. (2024) "Location-Based Service Application for Technician Allocation," International Journal of Computer Applications (IJCA), 2024.
6. Shukla S. G., et al. (2024) "Complaint Registration and Tracking System with Image Support," Journal of Emerging Technologies and Innovative Research (JETIR), 2024.
7. Singh P., et al. (2023) "Maintenance Management System using Cloud-Based Architecture," International Journal of Advanced Research in Computer Science (IJARCS), 2023.
8. Gupta S., et al. (2023) "Web-Based Complaint Handling and Tracking System," International Journal of Engineering Research & Technology (IJERT), 2023.
9. Walwadkar D., et al. (2022) "Smart Issue Reporting System with Real-Time Updates," International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), 2022.
10. Zhang Y., et al. (2021) "Location-Based Service Platforms for On-Demand Service Systems," IEEE Access, 2021.

